Data Handling and Visualization

Class Activity - 2

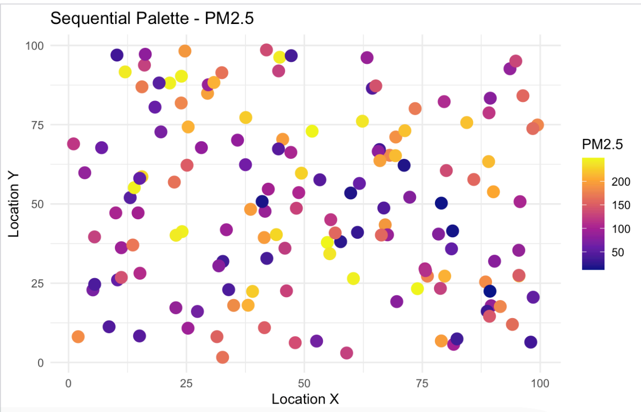
Date: 15/07/2026

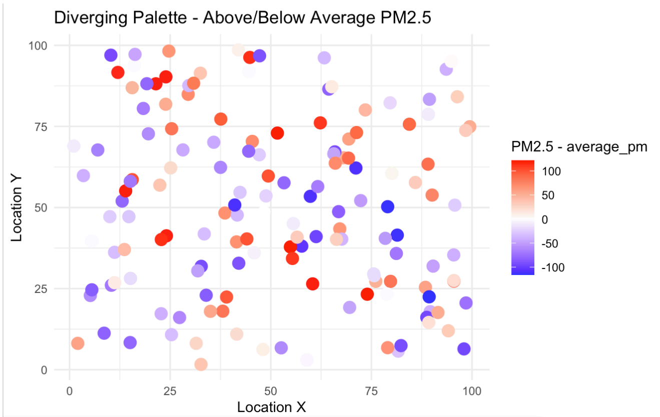
1. Smart City Air Quality Monitoring

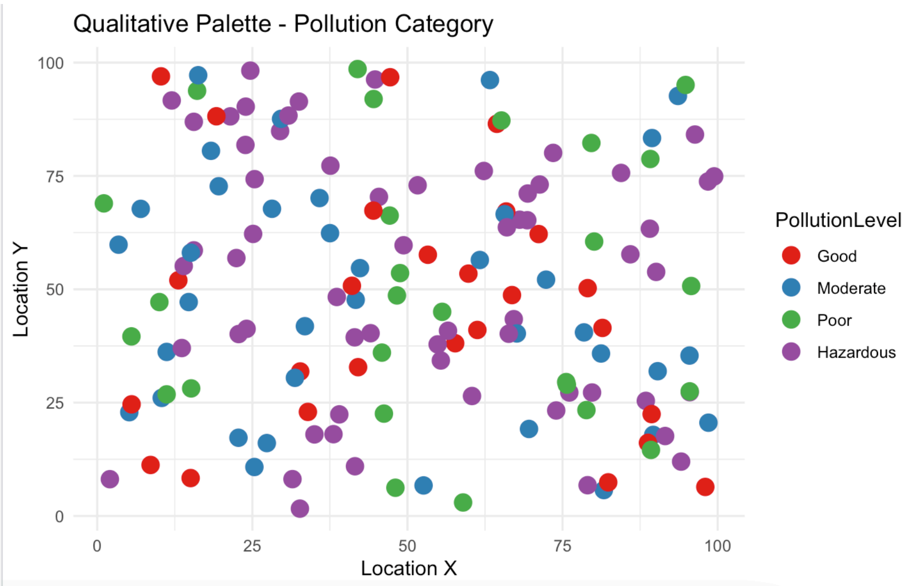
```
library(ggplot2)
library(dplyr)
library(viridis)

set.seed(123)

air_quality <- data.frame(
  SensorID = 1:150,
  X = runif(150, 1, 100),
  Y = runif(150, 1, 100),
  PM2.5 = runif(150, 10, 250),
  NO2 = runif(150, 5, 120),
  CO = runif(150, 0.5, 8),
  Temperature = runif(150, 18, 42),
  Humidity = runif(150, 30, 95)
)

head(air_quality)

ggplot(air_quality,
       aes(X, Y, color = PM2.5)) +
  geom_point(size = 4) +
  scale_color_viridis_c(option = "C") +
  labs(title = "Sequential Palette - PM2.5",
       x = "Location X",
       y = "Location Y") +
  theme_minimal()


average_pm <- mean(air_quality$PM2.5)

ggplot(air_quality,
       aes(X, Y,
           color = PM2.5 - average_pm)) +
  geom_point(size = 4) +
  scale_color_gradient2(
    low = "blue",
    mid = "white",
    high = "red",
    midpoint = 0
  ) +
  labs(title = "Diverging Palette - Above/Below Average PM2.5",
       x = "Location X",
       y = "Location Y") +
  theme_minimal()


air_quality$PollutionLevel <- cut(
  air_quality$PM2.5,
  breaks = c(0,50,100,150,250),
  labels = c("Good","Moderate","Poor","Hazardous")
)

ggplot(air_quality,
       aes(X, Y,
           color = PollutionLevel)) +
  geom_point(size = 4) +
  scale_color_brewer(palette = "Set1") +
```

```
labs(title = "Qualitative Palette - Pollution Category",
     x = "Location X",
     y = "Location Y") +
theme_minimal()
```

```
head(air_quality)
SensorID          X        Y     PM2.5       NO2       CO Temperature Humidity
       1 29.470174 84.89786 198.29806  65.23960 2.279223    24.66646 90.04045
       2 79.042208 50.25520  12.26318  45.22721 5.648678    23.98576 65.26889
       3 41.488715 39.40299 196.97581  32.66103 2.193638    21.82645 85.40370
       4 88.418723 25.39845 185.05376  11.69206 2.888709    19.71912 67.93159
       5 94.106261 11.99855 161.23164  32.21127 1.804879    31.37850 73.44104
       6  5.510093 39.60945 125.41860 107.35896 6.510722    29.75609 63.23545
```



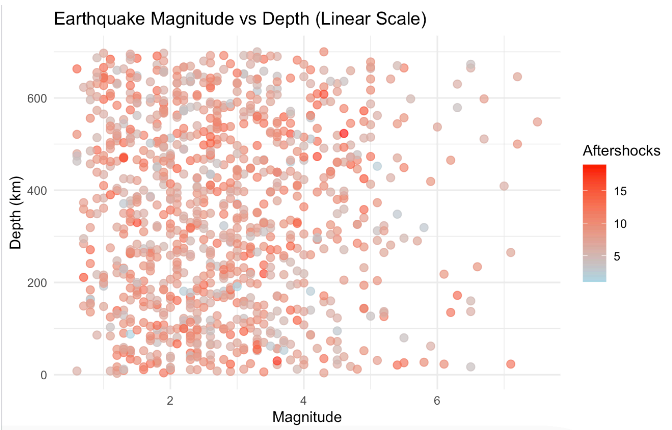




Question 2 – Earthquake Risk Analysis

```
library(ggplot2)
set.seed(123)
```

```
n <- 1000

earthquake <- data.frame(
  Magnitude = round(rbeta(n, 2, 6) * 9 + 0.5, 1),
  Depth = round(runif(n, 1, 700)),
  Latitude = runif(n, -90, 90),
  Longitude = runif(n, -180, 180),
  Aftershocks = rpois(n, lambda = 8)
)

head(earthquake)


ggplot(earthquake,
       aes(x = Magnitude,
           y = Depth,
           color = Aftershocks)) +
  geom_point(size = 2.5, alpha = 0.7) +
  scale_color_gradient(low = "lightblue",
                       high = "red") +
  labs(
    title = "Earthquake Magnitude vs Depth (Linear Scale)",
    x = "Magnitude",
    y = "Depth (km)",
    color = "Aftershocks"
```

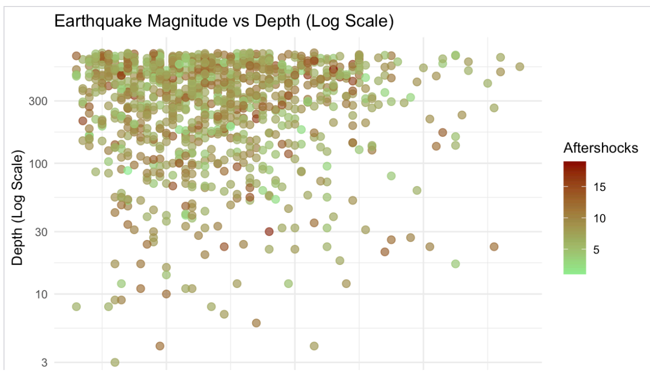


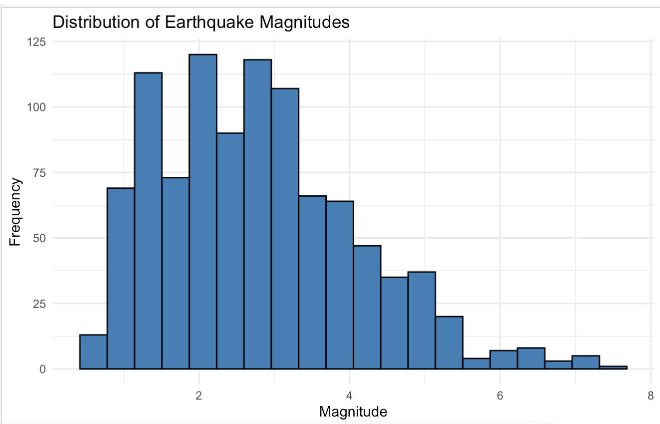

```
  ) +
  theme_minimal()


ggplot(earthquake,
       aes(x = Magnitude,
           y = Depth,
           color = Aftershocks)) +
  geom_point(size = 2.5, alpha = 0.7) +
  scale_y_log10() +
  scale_color_gradient(low = "lightgreen",
                       high = "darkred") +
  labs(
    title = "Earthquake Magnitude vs Depth (Log Scale)",
    x = "Magnitude",
    y = "Depth (Log Scale)",
    color = "Aftershocks"
  ) +
  theme_minimal()


ggplot(earthquake,
       aes(Magnitude)) +
```

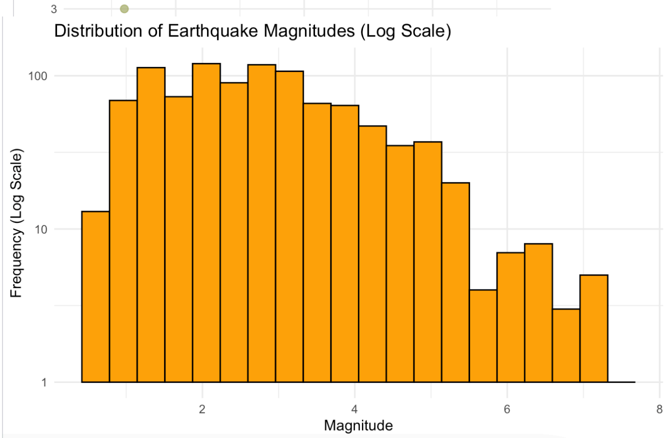
```r
  geom_histogram(
    bins = 20,
    fill = "steelblue",
    color = "black"
  ) +
  labs(
    title = "Distribution of Earthquake Magnitudes",
    x = "Magnitude",
    y = "Frequency"
  ) +
  theme_minimal()
```

```r
ggplot(earthquake,
       aes(Magnitude)) +
  geom_histogram(
    bins = 20,
    fill = "orange",
    color = "black"
  ) +
  scale_y_log10() +
  labs(
    title = "Distribution of Earthquake Magnitudes (Log Scale)",
    x = "Magnitude",
    y = "Frequency (Log Scale)"
  ) +
  theme_minimal()
```

```r
head(earthquake)
Magnitude Depth  Latitude   Longitude Aftershocks
      1.9   108 -11.75630 -177.815530          11
      2.4   182 -28.62917  158.691918           7
      6.3   172 -27.33660  -23.955254          13
      2.9   219 -35.24882  119.952450           8
      3.0   156  74.61287    5.486770          11
      3.6    22 -38.95581   -7.186249           7
```







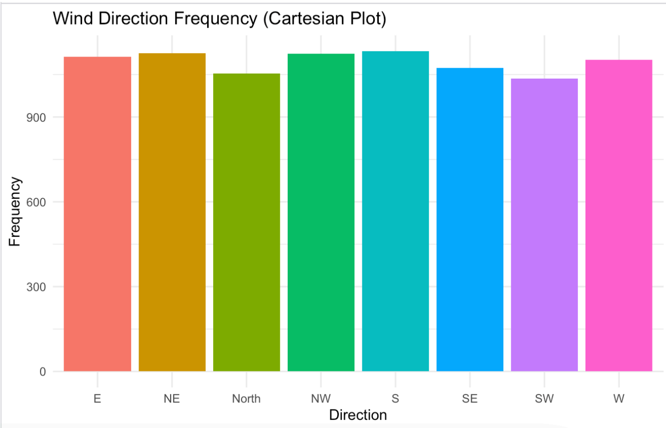
Question 3 – Wind Direction Analysis for Airport Operations

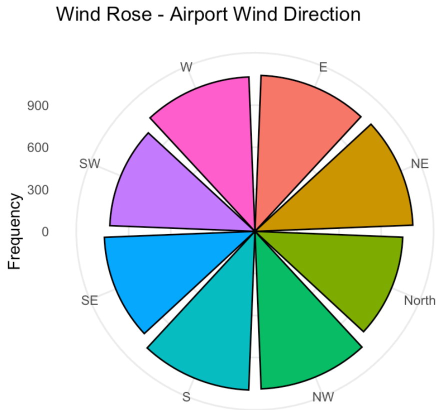
```
library(ggplot2)
library(dplyr)
set.seed(123)

n <- 365 * 24   # Hourly data for one year

wind_data <- data.frame(
  Hour = 1:n,
  WindSpeed = round(runif(n, 2, 30), 1),      # km/h
  WindDirection = round(runif(n, 0, 360)),    # Degrees
  Season = sample(c("Winter", "Spring", "Summer", "Monsoon"),
                  n, replace = TRUE)
)

head(wind_data)
wind_data$Direction <- cut(
  wind_data$WindDirection,
  breaks = c(-0.1,22.5,67.5,112.5,157.5,202.5,247.5,
           292.5,337.5,360),
  labels = c("N","NE","E","SE","S","SW","W","NW","N"),
  include.lowest = TRUE
)

# Remove duplicated "N"
wind_data$Direction <- as.character(wind_data$Direction)
wind_data$Direction[wind_data$Direction == "N"] <- "North"
ggplot(wind_data,
       aes(x = Direction,
          fill = Direction)) +
  geom_bar() +
  labs(
```

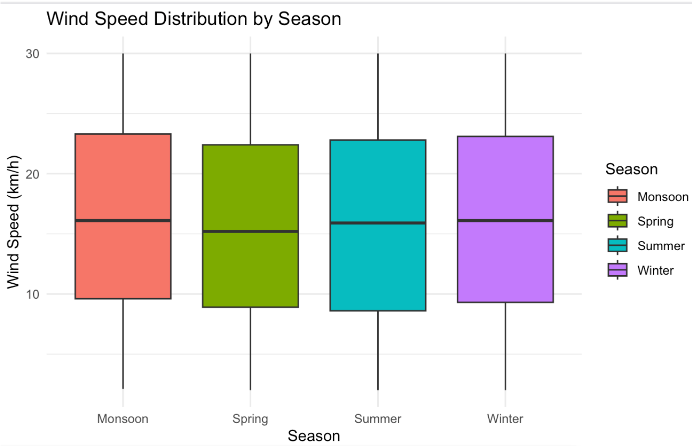
```
    title = "Wind Direction Frequency (Cartesian Plot)",
    x = "Direction",
    y = "Frequency"
  ) +
  theme_minimal() +
  theme(legend.position = “none")

ggplot(wind_data,
       aes(x = Direction,
           fill = Direction)) +
  geom_bar(color = "black") +
  coord_polar(start = 0) +
  labs(
    title = "Wind Rose - Airport Wind Direction",
    x = "",
    y = "Frequency"
  ) +
  theme_minimal() +
  theme(legend.position = "none")

ggplot(wind_data,
       aes(x = Direction,
           fill = Season)) +
  geom_bar(color = "black") +
  coord_polar() +
  facet_wrap(~Season) +
  labs(
    title = "Season-wise Wind Rose",
    x = "",
    y = "Frequency"
  ) +
  theme_minimal()

ggplot(wind_data,
       aes(x = Season,
           y = WindSpeed,
           fill = Season)) +
  geom_boxplot() +
```

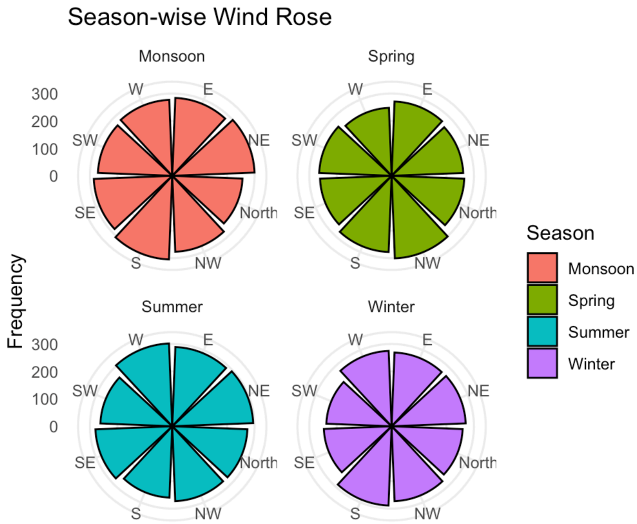
```r
labs(
  title = "Wind Speed Distribution by Season",
  x = "Season",
  y = "Wind Speed (km/h)"
) +
theme_minimal()
```

```
head(wind_data)
Hour WindSpeed WindDirection Season
   1      10.1            28 Summer
   2      24.1           168 Winter
   3      13.5            69 Spring
   4      26.7           353 Winter
   5      28.3           335 Summer
   6       3.3           187 Spring
```









Question 4 – Hospital Patient Monitoring Dashboard

```r
library(ggplot2)
library(dplyr)
library(patchwork)
set.seed(123)
n <- 200

patient_data <- data.frame(
```

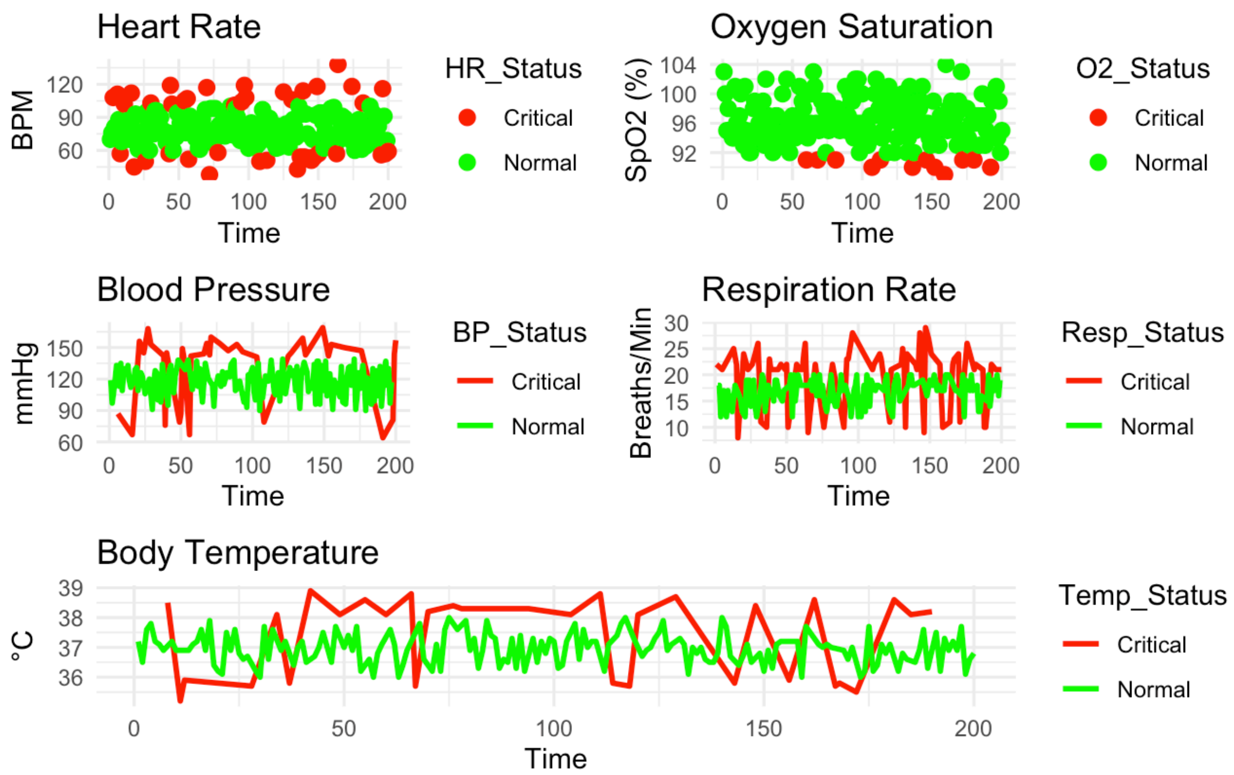
```
  Time = 1:n,
  HeartRate = round(rnorm(n, 80, 18)),
  Oxygen = round(rnorm(n, 96, 3)),
  SystolicBP = round(rnorm(n, 120, 20)),
  Respiration = round(rnorm(n, 18, 4)),
  Temperature = round(rnorm(n, 37, 0.7),1)
)
head(patient_data)
patient_data$HR_Status <- ifelse(
  patient_data$HeartRate < 60 | patient_data$HeartRate > 100,
  "Critical",
  "Normal"
)
patient_data$O2_Status <- ifelse(
  patient_data$Oxygen < 92,
  "Critical",
  "Normal"
)
patient_data$BP_Status <- ifelse(
  patient_data$SystolicBP < 90 | patient_data$SystolicBP > 140,
  "Critical",
  "Normal"
)
patient_data$Resp_Status <- ifelse(
  patient_data$Respiration < 12 | patient_data$Respiration > 20,
  "Critical",
  "Normal"
)
patient_data$Temp_Status <- ifelse(
  patient_data$Temperature < 36 | patient_data$Temperature > 38,
  "Critical",
  "Normal"
)
p1 <- ggplot(patient_data,
             aes(Time, HeartRate,
                 color = HR_Status)) +
  geom_point(size = 2.5) +
  scale_color_manual(values = c(
```

```
    "Normal" = "green",
    "Critical" = "red"
  )) +
  labs(title = "Heart Rate",
       x = "Time",
       y = "BPM") +
  theme_minimal()
p2 <- ggplot(patient_data,
             aes(Time, Oxygen,
                 color = O2_Status)) +
  geom_point(size = 2.5) +
  scale_color_manual(values = c(
    "Normal" = "green",
    "Critical" = "red"
  )) +
  labs(title = "Oxygen Saturation",
       x = "Time",
       y = "SpO2 (%)") +
  theme_minimal()

p3 <- ggplot(patient_data,
             aes(Time, SystolicBP,
                 color = BP_Status)) +
  geom_line(size = 1) +
  scale_color_manual(values = c(
    "Normal" = "green",
    "Critical" = "red"
  )) +
  labs(title = "Blood Pressure",
       x = "Time",
       y = "mmHg") +
  theme_minimal()
p4 <- ggplot(patient_data,
             aes(Time, Respiration,
                 color = Resp_Status)) +
  geom_line(size = 1) +
  scale_color_manual(values = c(
    "Normal" = "green",
```

```
  "Critical" = "red"
 )) +
 labs(title = "Respiration Rate",
      x = "Time",
      y = "Breaths/Min") +
 theme_minimal()

p5 <- ggplot(patient_data,
             aes(Time, Temperature,
                 color = Temp_Status)) +
 geom_line(size = 1) +
 scale_color_manual(values = c(
  "Normal" = "green",
  "Critical" = "red"
 )) +
 labs(title = "Body Temperature",
      x = "Time",
      y = "°C") +
 theme_minimal()
```

```
(p1 | p2) /
 (p3 | p4) /
 p5
```

```
head(patient_data)
Time HeartRate Oxygen SystolicBP Respiration Temperature
   1        70    103        119          22        37.2
   2        76    100         97          18        36.5
   3       108     95        107          18        37.6
   4        81     98        119          12        37.8
   5        82     95        133          21        37.2
   6       111     95         87          17        37.1
```



Question 5 – Global Climate Change Study

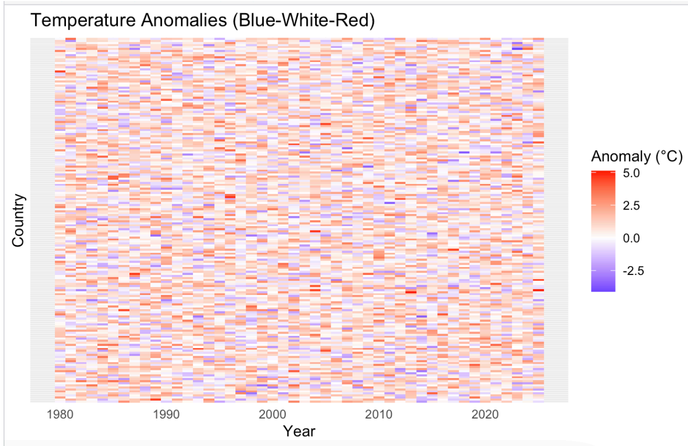
```
library(ggplot2)
set.seed(123)
countries <- paste("Country", 1:180)
years <- 1980:2025

climate_data <- expand.grid(
  Country = countries,
  Year = years
)

climate_data$TempAnomaly <- round(
  rnorm(nrow(climate_data), mean = 0.5, sd = 1.2),
  2
)
head(climate_data)
ggplot(climate_data,
       aes(x = Year,
           y = Country,
           fill = TempAnomaly)) +
  geom_tile() +
  scale_fill_gradient2(
    low = "blue",
    mid = "white",
    high = "red",
    midpoint = 0
  ) +
  labs(
    title = "Temperature Anomalies (Blue-White-Red)",
    fill = "Anomaly (°C)"
  ) +
  theme_minimal() +
  theme(
    axis.text.y = element_blank(),
    axis.ticks.y = element_blank()
  )
```

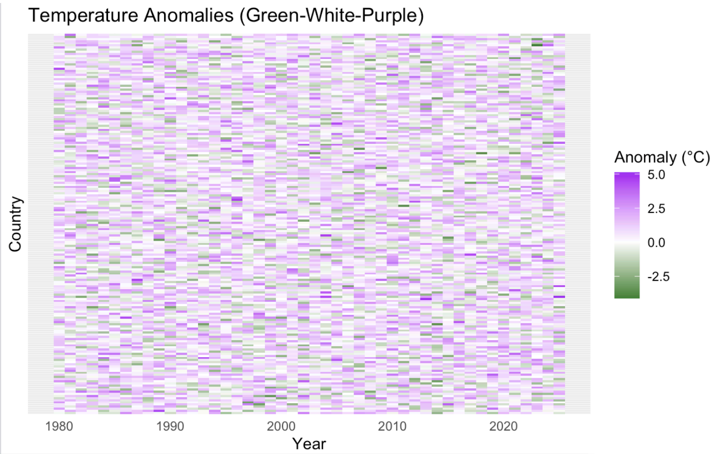
```
ggplot(climate_data,
       aes(x = Year,
           y = Country,
           fill = TempAnomaly)) +
  geom_tile() +
  scale_fill_gradient2(
    low = "darkgreen",
    mid = "white",
    high = "purple",
    midpoint = 0
  ) +
  labs(
    title = "Temperature Anomalies (Green-White-Purple)",
    fill = "Anomaly (°C)"
  ) +
  theme_minimal() +
  theme(
    axis.text.y = element_blank(),
    axis.ticks.y = element_blank()
  )
```

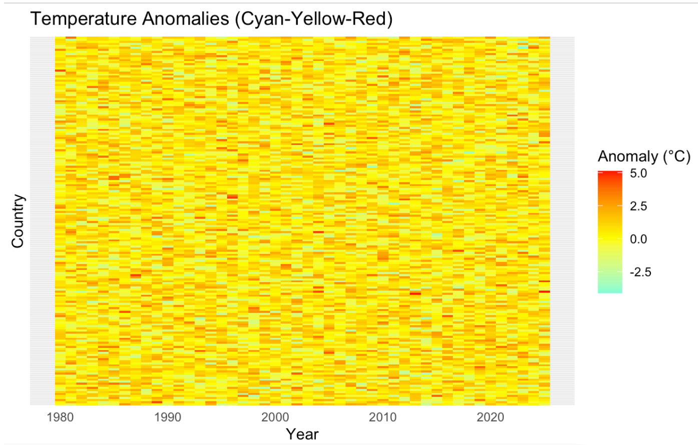
```
ggplot(climate_data,
       aes(x = Year,
           y = Country,
           fill = TempAnomaly)) +
  geom_tile() +
  scale_fill_gradient2(
    low = "cyan",
    mid = "yellow",
    high = "red",
    midpoint = 0
  ) +
  labs(
    title = "Temperature Anomalies (Cyan-Yellow-Red)",
    fill = "Anomaly (°C)"
  ) +
  theme_minimal() +
  theme(
```

```
  axis.text.y = element_blank(),
  axis.ticks.y = element_blank()
)
```

```
> head(climate_data)
    Country Year TempAnomaly
1 Country 1 1980       -0.17
2 Country 2 1980        0.22
3 Country 3 1980        2.37
4 Country 4 1980        0.58
5 Country 5 1980        0.66
6 Country 6 1980        2.56
```







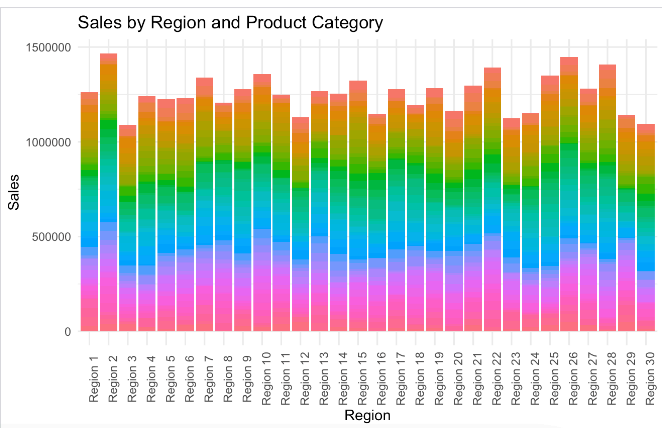
Question 6 – E-Commerce Sales Analytics

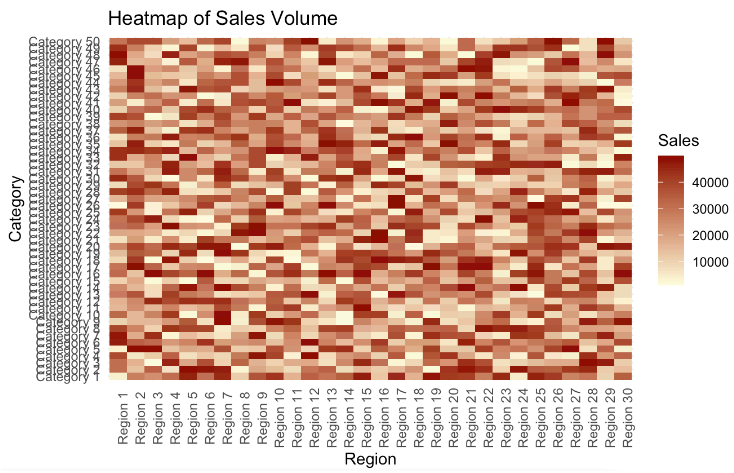
```
library(ggplot2)
set.seed(123)

categories <- paste("Category", 1:50)
regions <- paste("Region", 1:30)
sales_data <- expand.grid(
  Category = categories,
  Region = regions
)
sales_data$Sales <- sample(1000:50000,
                           nrow(sales_data),
                           replace = TRUE)
head(sales_data)
```

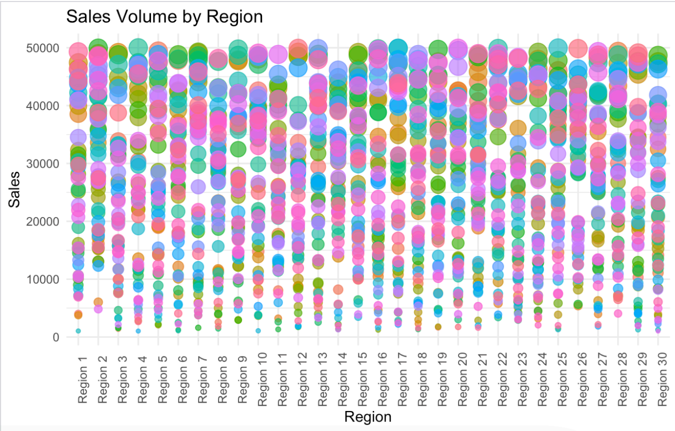
```
ggplot(sales_data,
       aes(x = Region,
           y = Sales,
           fill = Category)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Sales by Region and Product Category",
    x = "Region",
    y = "Sales"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 90),
    legend.position = "none"
  )
ggplot(sales_data,
       aes(x = Region,
           y = Category,
           fill = Sales)) +
  geom_tile() +
  scale_fill_gradient(
    low = "lightyellow",
    high = "darkred"
  ) +
  labs(
    title = "Heatmap of Sales Volume",
    fill = "Sales"
  ) +
  theme_minimal() +
  theme(
    axis.text.x = element_text(angle = 90)
  )
ggplot(sales_data,
       aes(x = Region,
           y = Sales,
           color = Category,
           size = Sales)) +
  geom_point(alpha = 0.7) +
```

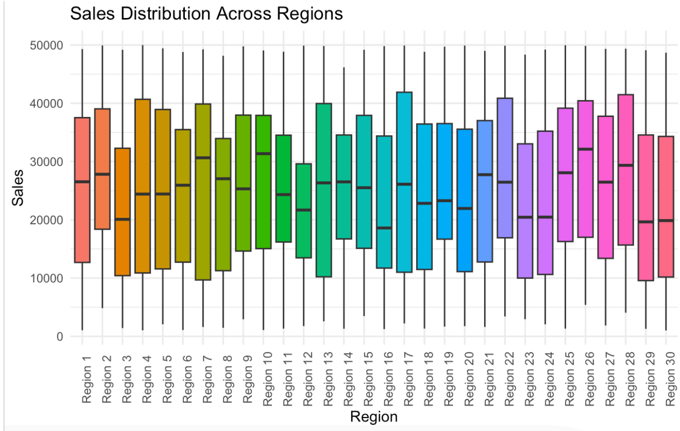
```
labs(
  title = "Sales Volume by Region",
  x = "Region",
  y = "Sales"
) +
theme_minimal() +
theme(
  axis.text.x = element_text(angle = 90),
  legend.position = "none"
)
ggplot(sales_data,
     aes(x = Region,
         y = Sales,
         fill = Region)) +
geom_boxplot() +
labs(
  title = "Sales Distribution Across Regions",
  x = "Region",
  y = "Sales"
) +
theme_minimal() +
theme(
  axis.text.x = element_text(angle = 90),
  legend.position = "none"
)
```

```
> head(sales_data)
    Category   Region Sales
1 Category 1 Region 1  3985
2 Category 2 Region 1 30924
3 Category 3 Region 1 30709
4 Category 4 Region 1 38528
5 Category 5 Region 1  3756
6 Category 6 Region 1 46403
```





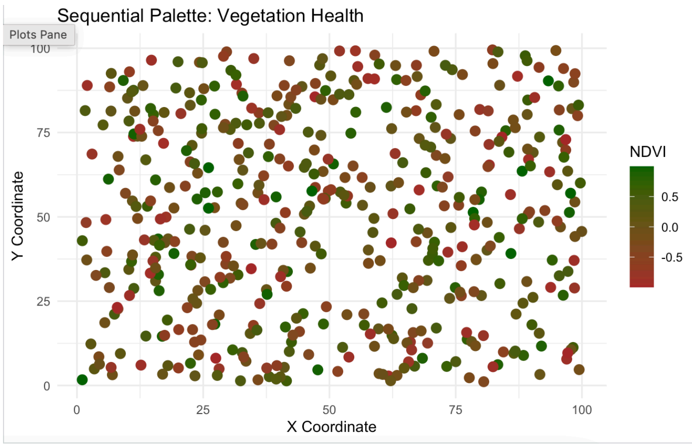




Question 7: Satellite Image Vegetation Analysis

```
library(ggplot2)
set.seed(123)
n <- 500
ndvi_data <- data.frame(
  X = runif(n, 1, 100),
  Y = runif(n, 1, 100),
  NDVI = runif(n, -1, 1)
)
head(ndvi_data)
ggplot(ndvi_data,
       aes(x = X,
           y = Y,
           color = NDVI)) +
  geom_point(size = 3) +
  scale_color_gradient(
    low = "brown",
    high = "darkgreen"
  ) +
  labs(
    title = "Sequential Palette: Vegetation Health",
    x = "X Coordinate",
    y = "Y Coordinate",
    color = "NDVI"
  ) +
  theme_minimal()
ggplot(ndvi_data,
       aes(x = X,
           y = Y,
```

```
> head(ndvi_data)
          X         Y        NDVI
1 29.470174 36.007002 -0.45275453
2 79.042208 37.277703  0.18773387
3 41.488715 29.422913 -0.67963037
4 88.418723  8.917318  0.70686048
5 94.106261 37.179973  0.69547832
6  5.510093 18.623368 -0.04422637
>
```

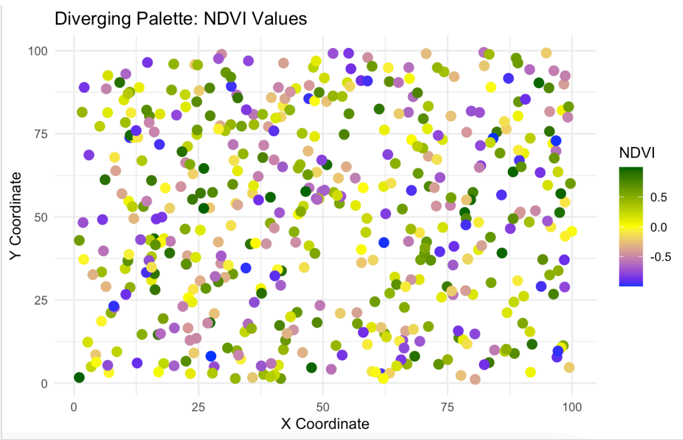


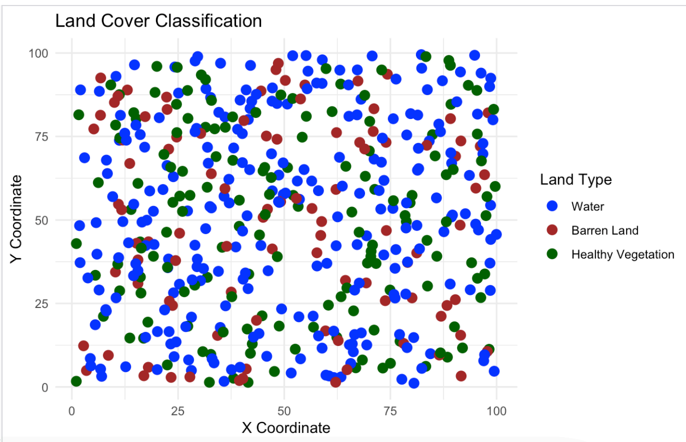

```
          color = NDVI)) +
  geom_point(size = 3) +
  scale_color_gradient2(
    low = "blue",
    mid = "yellow",
    high = "darkgreen",
    midpoint = 0
  ) +
  labs(
    title = "Diverging Palette: NDVI Values",
    x = "X Coordinate",
    y = "Y Coordinate",
    color = "NDVI"
  ) +
  theme_minimal()
ndvi_data$LandType <- cut(
  ndvi_data$NDVI,
  breaks = c(-1, 0, 0.4, 1),
  labels = c("Water", "Barren Land", "Healthy Vegetation")
)
ggplot(ndvi_data,
       aes(x = X,
          y = Y,
          color = LandType)) +
  geom_point(size = 3) +
  scale_color_manual(values = c(
    "Water" = "blue",
    "Barren Land" = "brown",
    "Healthy Vegetation" = "darkgreen"
```

```
)) +
labs(
  title = "Land Cover Classification",
  x = "X Coordinate",
  y = "Y Coordinate",
  color = "Land Type"
) +
theme_minimal()
```

Question 8 – Cryptocurrency Market Volatility

```
library(ggplot2)
set.seed(123)
dates <- seq(as.Date("2022-01-01"), by = "day", length.out = 365)
crypto_data <- data.frame(
  Date = rep(dates, 4),
  Cryptocurrency = rep(c("Bitcoin", "Ethereum", "Solana", "Cardano"),
                       each = 365),
  Price = c(
```

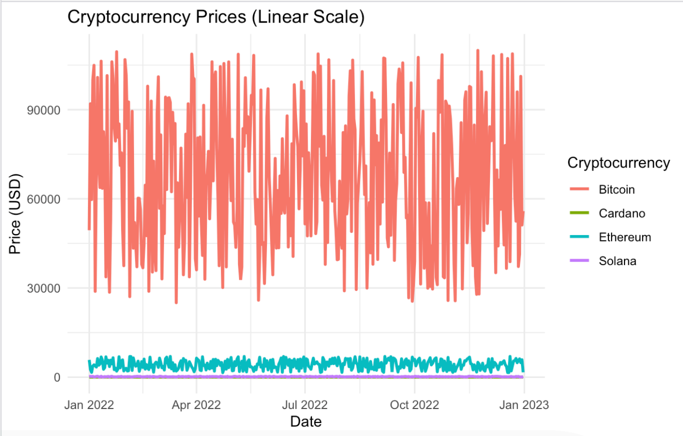


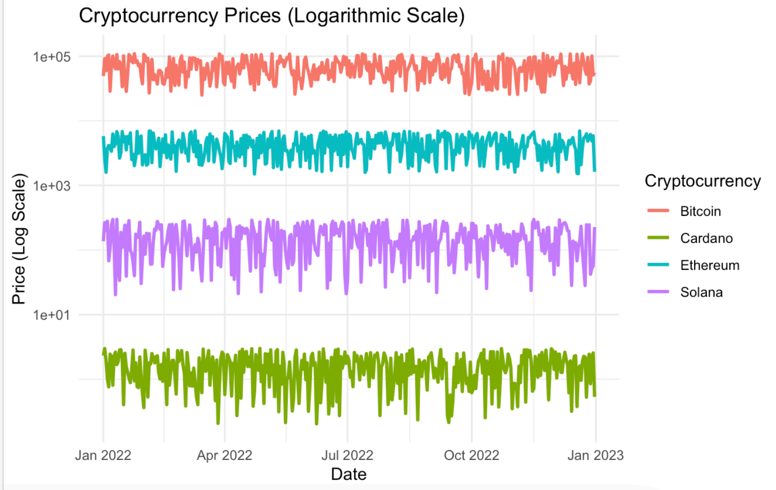


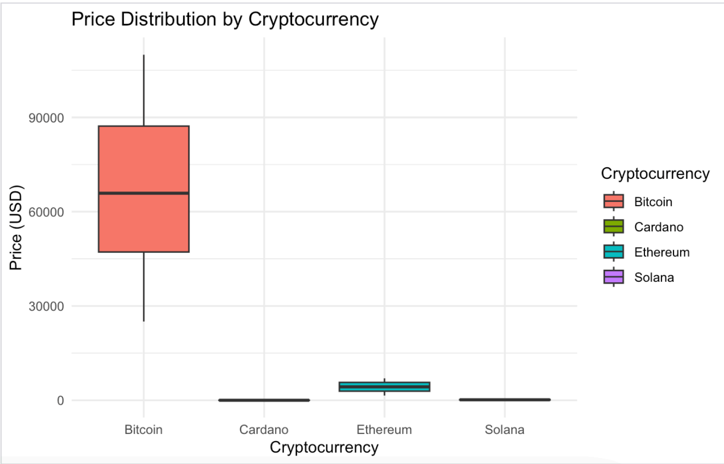

```
    runif(365, 25000, 110000),
    runif(365, 1500, 7000),
    runif(365, 20, 300),
    runif(365, 0.2, 3)
  )
)
head(crypto_data)
ggplot(crypto_data,
    aes(x = Date,
        y = Price,
        color = Cryptocurrency)) +
  geom_line(size = 1) +
  labs(
    title = "Cryptocurrency Prices (Linear Scale)",
```

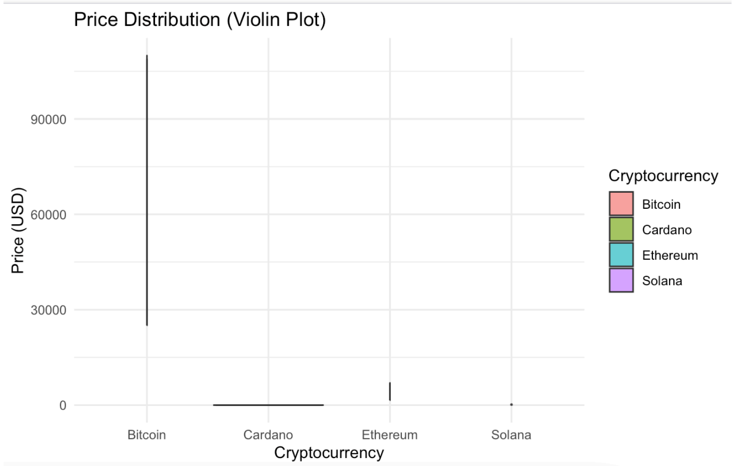
```
    x = "Date",
    y = "Price (USD)"
  ) +
  theme_minimal()
ggplot(crypto_data,
       aes(x = Date,
           y = Price,
           color = Cryptocurrency)) +
  geom_line(size = 1) +
  scale_y_log10() +
  labs(
    title = "Cryptocurrency Prices (Logarithmic Scale)",
    x = "Date",
    y = "Price (Log Scale)"
  ) +
  theme_minimal()
ggplot(crypto_data,
       aes(x = Cryptocurrency,
           y = Price,
           fill = Cryptocurrency)) +
  geom_boxplot() +
  labs(
    title = "Price Distribution by Cryptocurrency",
    x = "Cryptocurrency",
    y = "Price (USD)"
  ) +
  theme_minimal()
ggplot(crypto_data,
       aes(x = Cryptocurrency,
           y = Price,
           fill = Cryptocurrency)) +
  geom_violin(alpha = 0.7) +
  labs(
    title = "Price Distribution (Violin Plot)",
    x = "Cryptocurrency",
    y = "Price (USD)"
  ) +
  theme_minimal()
```

```
> head(crypto_data)
        Date Cryptocurrency     Price
1 2022-01-01        Bitcoin  49444.09
2 2022-01-02        Bitcoin  92005.94
3 2022-01-03        Bitcoin  59763.04
4 2022-01-04        Bitcoin 100056.48
5 2022-01-05        Bitcoin 104939.72
6 2022-01-06        Bitcoin  28872.30
```









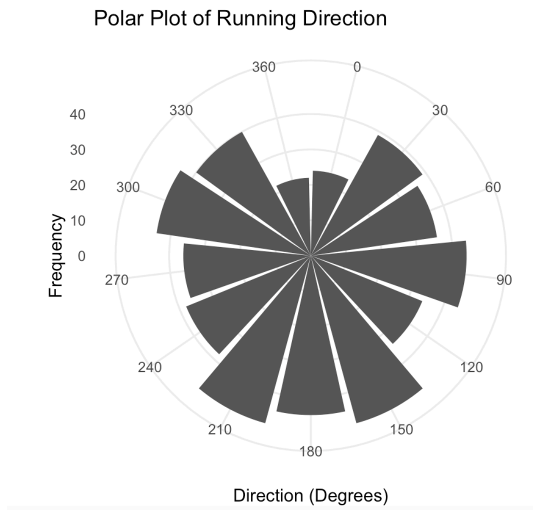
Question 9 – Marathon Runner Performance Analysis

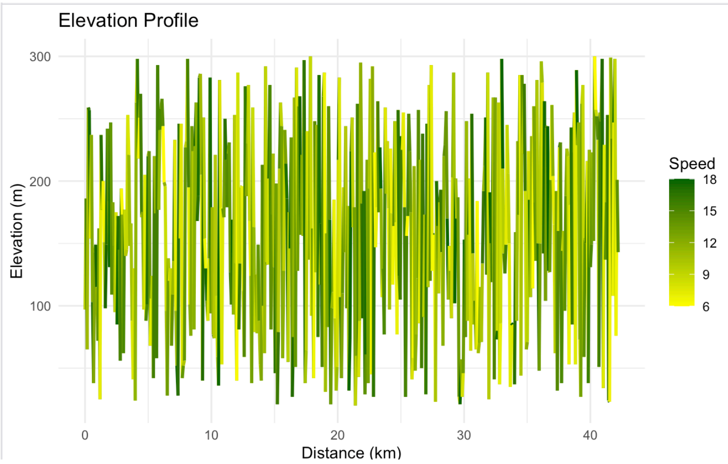
```
library(ggplot2)
set.seed(123)
n <- 500
runner_data <- data.frame(
  Distance = seq(0, 42.2, length.out = n),
  Speed = round(runif(n, 6, 18), 1),
  HeartRate = round(runif(n, 110, 190)),
  Elevation = round(runif(n, 20, 300)),
  Direction = runif(n, 0, 360)
)
head(runner_data)

ggplot(runner_data,
       aes(x = Distance,
          y = Speed,
          color = Speed)) +
  geom_line(size = 1) +
  scale_color_gradient(
    low = "blue",
    high = "red"
  ) +
```

```
  labs(
    title = "Runner Speed Throughout the Marathon",
    x = "Distance (km)",
    y = "Speed (km/h)",
    color = "Speed"
  ) +
  theme_minimal()

ggplot(runner_data,
       aes(x = factor(round(Direction/30) * 30),
           fill = Speed)) +
  geom_bar() +
  coord_polar() +
  scale_fill_gradient(
    low = "lightblue",
    high = "darkred"
  ) +
  labs(
    title = "Polar Plot of Running Direction",
    x = "Direction (Degrees)",
    y = "Frequency",
    fill = "Speed"
  ) +
  theme_minimal()

ggplot(runner_data,
       aes(x = Distance,
           y = HeartRate,
           color = Speed)) +
  geom_line(size = 1) +
  scale_color_gradient(
    low = "green",
    high = "red"
  ) +
  labs(
    title = "Heart Rate During Marathon",
    x = "Distance (km)",
    y = "Heart Rate (BPM)",
    color = "Speed"
  ) +
  theme_minimal()

ggplot(runner_data,
       aes(x = Distance,
           y = Elevation,
           color = Speed)) +
  geom_line(size = 1) +
  scale_color_gradient(
```

```
  low = "yellow",
  high = "darkgreen"
) +
labs(
  title = "Elevation Profile",
  x = "Distance (km)",
  y = "Elevation (m)",
  color = "Speed"
) +
theme_minimal()
```

```
> head(runner_data)
    Distance Speed HeartRate Elevation Direction
1 0.00000000   9.5       138        97 337.69017
2 0.08456914  15.5       139       186 355.68119
3 0.16913828  10.9       133        65 164.27504
4 0.25370741  16.6       116       259  83.02138
5 0.33827655  17.3       139       257 250.37614
6 0.42284569   6.5       124       154 200.38764
```





Question 10 – COVID-19 Vaccination Campaign Dashboard

```
library(ggplot2)
set.seed(123)
districts <- paste("District", 1:20)
age_groups <- c("18-30", "31-45", "46-60", "60+")
vaccine_types <- c("Covishield", "Covaxin", "Sputnik V")

vaccination_data <- expand.grid(
```

```
  District = districts,
  AgeGroup = age_groups,
  VaccineType = vaccine_types
)

vaccination_data$VaccinationRate <- sample(
  50:100,
  nrow(vaccination_data),
  replace = TRUE
)
head(vaccination_data)
ggplot(vaccination_data,
       aes(x = District,
           y = VaccinationRate,
           fill = VaccineType)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(
    title = "Vaccination Rates by District and Vaccine Type",
    x = "District",
    y = "Vaccination Rate (%)"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 90))

ggplot(vaccination_data,
       aes(x = District,
           y = AgeGroup,
           fill = VaccinationRate)) +
  geom_tile() +
  scale_fill_gradient(
    low = "lightyellow",
    high = "darkgreen"
  ) +
  labs(
    title = "Vaccination Heatmap",
    x = "District",
    y = "Age Group",
    fill = "Rate (%)"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 90))
vaccination_data$Status <- ifelse(
  vaccination_data$VaccinationRate >= 80,
  "High",
  "Low"
)
ggplot(vaccination_data,
       aes(x = District,
```

```
        y = VaccinationRate,
        color = Status)) +
  geom_point(size = 3) +
  scale_color_manual(values = c(
    "High" = "green",
    "Low" = "red"
  )) +
  labs(
    title = "Vaccination Status by District",
    x = "District",
    y = "Vaccination Rate (%)"
  ) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 90))
ggplot(vaccination_data,
      aes(x = VaccineType,
        y = VaccinationRate,
        fill = VaccineType)) +
  geom_boxplot() +
  labs(
    title = "Vaccination Rate Distribution by Vaccine Type",
    x = "Vaccine Type",
    y = "Vaccination Rate (%)"
  ) +
  theme_minimal()
```